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Mainstreaming Vulnerability in Adaptation Finance: Towards a Theory-Informed Analysis of Official Development Assistance in the World Economy

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Received: 11-09-2025 **Revised:** 12-31-2025 **Accepted:** 01-27-2026

Citation: Oskay, C. & Yardımcı, P. (2026). Mainstreaming vulnerability in adaptation finance: Towards a theory-informed analysis of Official Development Assistance in the world economy. *Chall. Sustain.*, *14*(2), 222–241. https://doi.org/10.56578/cis140201.

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Abstract: This study presented a theory-informed bibliometric review that explored the intersection of adaptation finance, vulnerability, and development cooperation within the climate finance literature. Anchored in the vulnerability-resilience framework, the study aims to map the conceptually-aligned financial models on adaptation, particularly how policy-driven instruments such as Official Development Assistance (ODA) have evolved within the world economy and debates about global macroeconomic policy. Utilizing a conceptually integrated search strategy, the analysis combined bibliographic coupling, thematic clustering, and theory-informed mapping techniques. The findings revealed that although adaptation-related concepts held a central place in global policy frameworks (e.g., Sustainable Development Goals (SDGs) 13 and 17), their representations in the academic literature remained uneven and fragmented. Structural clusters reflected the dominance of Global North institutions and mitigation-centered research whereas emerging thematic patterns indicated growing emphasis on context-specific and vulnerability-sensitive adaptation finance. Comparative insights from sectoral ODA data confirmed the thematic gaps identified in the bibliometric analysis and underscored the persistent disconnect between financial flows and local adaptation needs. By linking bibliometric insights with patterns of institutional finance, this study offered an integrative perspective on climate-oriented development and contributed to the agenda of global economic transformation. In doing so, it addressed a significant research gap via combining integrated theory-driven bibliometric mapping with analysis of policy-centered development finance.

Keywords: Adaptation finance; Vulnerability; Official Development Assistance; World economy; Macroeconomic policy; Bibliometric analysis; Sustainable development

1. Introduction

The global economy is marked by profound inequalities in both the impacts of climate change and the distribution of financial responses. This study investigated how adaptation finance had been institutionalized at the global level, the thematic contours around which it had evolved, and how the structural vulnerabilities of developing countries were represented in both academic literature and policy frameworks. In doing so, it framed the interaction between climate adaptation and development finance within a theoretical and applied perspective rooted in the global political economy. Despite growing attention to adaptation, market-based finance mechanisms remain structurally limited in their capacity to reduce vulnerability, especially in the Global South. While frameworks such as the Paris Agreement and Sustainable Development Goals (SDGs) 13 and 17 promote inclusive finance, their conceptual reflection in scholarly discourse remains uneven and limited.

This study grounded its analysis in the vulnerability-resilience perspective pioneered by Adger (2006) which conceptualized climate vulnerability as a socially and institutionally embedded condition rather than a purely biophysical exposure. Adaptive capacity is shaped by structural factors such as institutional quality, socio-economic inequality, and governance coherence as this shift attention away from technocratic risk assessments toward the social and institutional drivers that determine how countries absorb and reorganize under climate stress. This perspective, increasingly emphasized across the successive Intergovernmental Panel on Climate Change

(IPCC) reports, frames adaptation as a development-oriented and context-specific process. In line with this shift, the Organisation for Economic Co-operation and Development (OECD) has operationalized vulnerability-sensitive adaptation through Official Development Assistance (ODA) instruments, such as the Rio Markers. These instruments link finance allocation to principles of local ownership, institutional alignment, and development effectiveness. Unlike mitigation finance which is mobilized largely through market-based mechanisms such as carbon trading and green investment, adaptation requires concessional and non-market public resources capable of addressing diverse and asymmetric vulnerabilities across the Global South. This divergence between market-centered mitigation finance and context-sensitive adaptation support underscores why ODA remains the primary vehicle for channeling resources to countries with limited institutional capacity and high structural fragility.

Although there is a growing policy shift toward adaptation within climate finance, it is difficult to determine the extent to which this transformation is reflected in the literature through a focus on vulnerability sensitivity and adaptive capacity. Yet, empirical studies such as those by Jain & Bardhan (2022) and Rauniyar et al. (2025) emphasized that for adaptation finance to effectively reduce vulnerability, it should support not only needs-based allocation but also structural factors such as adaptation readiness and institutional capacity. This highlights a visibility gap in the literature, where the concept of adaptive capacity has yet to be fully integrated, both theoretically and representationally. The theoretical innovation of this study is the integration of vulnerability-mainstreaming theory with empirical bibliometric mapping and ODA policy analysis. While earlier research tended to examine adaptation finance or vulnerability theory in isolation, this study combined these two domains to illustrate how knowledge production via bibliometric patterns and financial governance structures (ODA allocation mechanisms) jointly shaped the institutionalization of adaptation finance.

This integrated view enables a theory-informed interpretation of bibliometric clusters, to reveal whether and how vulnerability-sensitive concepts enter or fail to enter the agenda of global adaptation finance. By linking scholarly discourse with policy architecture, the study provided a novel framework for assessing the coherence, gaps, and misalignments between academic debates and real-world priorities of adaptation finance. Accordingly, this study positioned adaptation finance both as a reflection of the institutionalization of vulnerability within development policy and as an indicator of a paradigmatic shift from carbon-centered mitigation finance to equity-oriented adaptation finance. To assess the extent to which this transformation is represented in academic discourse, the study adopted a bibliometric approach based on thematic mapping. The bibliometric findings of this study revealed that vulnerability-sensitive adaptation finance was becoming increasingly visible in the literature; however, this visibility remained structurally limited. Furthermore, as corroborated by ODA data, significant disconnects persisted between financial flows and local adaptation needs.

The aim of this study is to identify and visualize the thematic structure of adaptation finance by employing a bibliometric approach that links key concepts such as ODA, governance, and resilience. Its contribution lies in integrating this theoretical framework with empirically bibliometric evidence, thereby illustrating how development finance is evolving toward more adaptive and equity-oriented paradigms. This framework also addresses the gap between the rising policy relevance of vulnerability-centered finance and its relatively limited representation in the academic literature.

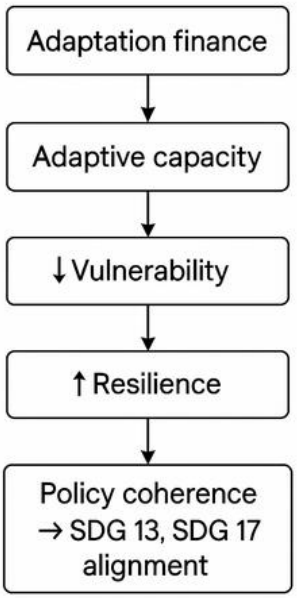

Figure 1. Theoretical pathway linking vulnerability, adaptive capacity, and adaptation finance

Figure 1 illustrates the conceptual pathway underpinning this study. It shows how vulnerability-mainstreaming, adaptive capacity, and ODA governance structures interact to shape the outcomes of adaptation finance. By linking bibliometric themes with institutional mechanisms, this framework clarifies the theoretical logic guiding the empirical analysis. Building on this conceptual model, the study addressed four research questions that examined (1) how vulnerability-sensitive concepts evolved in the literature of adaptation finance; (2) how ODA-based adaptation finance was thematically structured; (3) where governance and institutional gaps emerged; and (4) how these patterns aligned with the policy agendas of SDG 13 or 17.

2. Theoretical Framework and Positioning of the Literature

Responses to climate change are commonly classified into mitigation and adaptation; yet as emphasized by Smit et al. (2000), adaptation fundamentally concerned the capability of ecological, social, and economic systems to adjust internally to ongoing or anticipated climatic stresses rather than external emission-reducing interventions. Building on this distinction, Smit & Wandel (2006) conceptualized vulnerability as a dynamic outcome of the interaction among exposure, sensitivity, and adaptive capacity. Turner et al. (2003) further demonstrated that exposure and sensitivity varied substantially across socio-institutional and biophysical contexts, thus shaping how climate impacts were experienced in specific settings. Within this framework, adaptive capacity emerged as the pivotal determinant of vulnerability reduction, drawing on multidimensional factors such as economic resources, institutional arrangements, governance quality, social capital, and technological access. Adaptation policy represents the operational expression of this capacity, which translates systemic strengths into strategies and practices aimed at reducing climate-related risks. Together, the concepts of vulnerability, adaptive capacity, and adaptation provide an integrated analytical foundation for understanding how societies respond to climate hazards.

The widely adopted formulation of vulnerability with functions of exposure, sensitivity, and adaptive capacity provides a dynamic basis for understanding how climate risks operate across socio-institutional contexts (Smit & Wandel, 2006). Within this structure, adaptive capacity constitutes the most influential component since communities facing similar climatic stresses exhibit markedly different vulnerability levels, depending on their capacity to anticipate, absorb, and reorganize under pressure. As emphasized by Smit & Pilifosova (2003), adaptive capacity extended beyond short-term response and included sustained processes of learning, flexibility, and transformation. This capacity is closely linked to resilience: while adaptive capacity enables systems to adjust, resilience ensures that these adjustments maintain long-term continuity (Nelson et al., 2007). In developing countries, external public finance, especially ODA, plays a crucial role in strengthening this institutional and socioeconomic capacity, rendering it an essential instrument for reducing structural vulnerability.

Resilience has been conceptualized as the ability to cope with shocks and maintain functional stability across ecological and social systems, whether through ecological thresholds (Holling, 1973) or social mechanisms that address governance and economic uncertainty (Adger, 2000). These perspectives converge in highlighting that resilience and adaptive capacity are mutually reinforcing determinants of vulnerability reduction. This relationship underpins the rationale for vulnerability-based adaptation finance, which views adaptation not merely as a technical intervention but as an institutional transformation that strengthens policy coherence and aligns with global objectives such as SDG 13 and SDG 17. Grounded in Adger (2006)'s vulnerability–resilience framework, this study adopted the view that enhancing adaptive capacity provided a pathway for integrating vulnerability reduction into sustainable development, particularly in contexts where ODA could redirect financial resources toward areas most critical for long-term climate resilience.

2.1 Mainstreaming Vulnerability in Adaptation Finance: A Conceptual Shift

Adger's approach to vulnerability provides a multidimensional perspective that conceptualizes climate risks not merely as a function of exposure but as a systemic issue intertwined with social inequalities, institutional capacity, and governance structures. Within this framework, vulnerability is shaped not only by exposure and sensitivity but also by the ability of a system to learn, adjust, and reorganize under climate stress, i.e., its adaptive capacity. The sustainability of this capacity is widely recognized as a key determinant of long-term vulnerability reduction (Smit & Pilifosova, 2003). Accordingly, adaptation policies should extend beyond technical measures and be mainstreamed into institutional arrangements and governance systems. In this study, mainstreaming was employed as an analytical lens that captured both the institutionalization of concepts such as vulnerability, adaptive capacity, and governance in the literature, and the shift of development finance instruments, particularly ODA, toward adaptation-oriented priorities. Since vulnerability reflects not only exposure but also the distribution of adaptive capacity across countries (Adger, 2006), contexts with limited capacity face deeper and more persistent climate risks. A vulnerability-oriented financing approach therefore provides a necessary policy pathway that acknowledges these structural disadvantages. This framework conceptually anchors the emerging paradigm of vulnerability-sensitive adaptation finance.

Since the early 2000s, research on the economics of adaptation has largely focused on the costs of climate

impacts and how they might be financed, often reducing adaptation to a technical or investment measure and overlooking critical social and institutional dimensions (Tol et al., 1998). Entry of adaptation into the agenda of the United Nations Framework Convention on Climate Change (UNFCCC) in the mid 1990s shifted the debate toward a broader understanding that recognized vulnerability, governance, and inequality as central to climate risk. This evolution was reflected in the IPCC Second Assessment Report, which identified sensitivity and adaptive capacity as key determinants of vulnerability (IPCC, 1996), and in subsequent assessments conceptualizing vulnerability through exposure, sensitivity, and the capacity to respond to climatic stress (IPCC, 2001). Adger (2006)'s influential framework further established vulnerability in social and institutional structures; he argued that limited institutional capacity, entrenched inequalities, and governance constraints shaped how societies experienced and responded to climate risks. Together, these contributions established the basis for understanding adaptation as a systemic, rather than solely technical, process.

Building on these foundations, the IPCC Fourth Assessment Report defined adaptation as an ongoing process through which social and ecological systems reorganized in response to actual or expected climate impacts, a process especially challenging in contexts marked by poverty, weak institutions, and scarcity of resources (IPCC, 2007). Misaligned interventions in development might further deepen vulnerabilities, thus demonstrating the need for policy coherence between climate and development planning (Tearfund, 2006). In parallel, the OECD frameworks began integrating adaptation into development cooperation, recognizing that managing climate risks requires strengthening institutional capacity and concerns about mainstreaming vulnerability across development policies (OECD, 2001). This institutional shift laid the groundwork for ODA as a primary mechanism supporting adaptation, with Persson & Klein (2009) noting that the growing emphasis on vulnerability-sensitive finance represented a structural evolution in global development assistance. Collectively, these developments highlighted a transition from viewing adaptation as a set of technical adjustments toward understanding it as a core component of sustainable development and long-term resilience building.

In parallel with this shift, ODA has been restructured under sustainable development principles to address poverty, environmental sustainability, and climate risks through multisectoral approaches. This evolution reframed adaptation from a secondary environmental issue to a development-oriented intervention embedded in strategy and implementation cycles, thus aligning ODA with climate-resilient and inclusive development goals (Persson & Klein, 2009). Nevertheless, vulnerability-focused adaptation remains conceptually and methodologically underrepresented in the academic literature.

2.2 Research Gaps and Analytical Justifications

Despite the growing emphasis on adaptation in policy, the academic literature continues to treat adaptation as a predominantly technical or project-based intervention rather than a socially and institutionally embedded process. Sherman et al. (2016) showed that adaptation had not been transformatively mainstreamed into development policy, with many "mainstreaming" initiatives merely repackaging existing development practices without addressing climate-specific risks. Similarly, Smit & Wandel (2006) emphasized that adaptation could not be understood solely through technical or economic measures; rather, it depended on multidimensional drivers such as social relations, governance systems, and institutional capacity.

A related conceptual gap stems from the limited integration of vulnerability-based approaches. While Eriksen et al. (2021) demonstrated that poorly designed adaptation projects could reinforce structural inequalities and generate maladaptation, the literature had not systematically explored how vulnerability, adaptive capacity, and institutional readiness shaped the allocation of finance. Bibliometric studies further highlighted that mitigation-centered themes dominated the research on climate finance, while vulnerability, adaptive capacity, and resilience remained marginalized (Mohamad, 2025). Although Adger (2006)'s vulnerability framework has been widely cited, its implications for adaptation finance flows, especially ODA, remain under-theorized and insufficiently mapped. Together, these gaps reveal a fragmented conceptual landscape in which adaptation finance is insufficiently linked to the theoretical constructs that define vulnerability, thus creating the need for a systematic framework that captures how vulnerability-oriented concepts are represented in scholarly discourse.

Existing bibliometric studies on climate finance demonstrated a strong mitigation bias, with analytical attention concentrated on themes such as green bonds, carbon markets, and environmental, social, and governance (ESG)-oriented investment instruments, hence leaving the intersection of adaptation finance, ODA, governance, and policy coherence largely unmapped. This imbalance limits the field in examining how vulnerability, adaptive capacity, and institutional readiness are reflected in adaptation-related scholarship. The representational asymmetry between major academic databases further reinforces this gap: while a Scopus-based query yields more than 370 records linking vulnerability to governance and adaptation finance, the Web of Science (WoS) contains only 1 relevant publication, indicating a structural fragmentation in the academic literature. When viewed together, these methodological shortcomings underscore the absence of a comprehensive bibliometric framework, which could capture how adaptation finance, in particular ODA-linked mechanisms, is positioned within broader climate policy debates.

Empirical research on adaptation finance revealed that effective adaptation was constrained not only by technical feasibility but also by social, institutional, and economic barriers. Adger et al. (2007) demonstrated that governance capacity, inequality, and financial limitations significantly shaped the viability of adaptation measures, with fiscal constraints affecting both low-income countries and local institutions. Although global climate policy frameworks increasingly prioritize allocating funds to the most vulnerable countries, as reflected in mechanisms such as the Green Climate Fund (Muccione et al., 2016), market-based instruments remain insufficient for channeling resources toward these contexts. Stoll et al. (2021) attributed the limited private-sector engagement in adaptation to market failures, thus underscoring the continued need for publicly supported and non-market-based mechanisms such as ODA.

At the same time, the effectiveness of ODA depends not only on the volume of finance but also on the adaptive capacity and institutional readiness of recipient countries. Evidence from Jain & Bardhan (2022) and Rauniyar et al. (2025) illustrated that although vulnerable countries often received more ODA, these funds did not consistently reduce vulnerability when governance capacity was weak or when allocations were misaligned with structural needs. This gap is further widened by a disconnect between policy and scholarship: while multilateral frameworks increasingly emphasize vulnerability-sensitive financing, academic research has yet to fully capture or analyze this shift.

Against these conceptual, methodological, and empirical gaps, this study provided an integrative analytical framework that systematically mapped how vulnerability-centered themes such as adaptive capacity, governance, and resilience were represented in the literature of adaptation finance. By linking these conceptual elements to ODA-based development finance and to policy coherence objectives reflected in SDG 13 and SDG 17, the study situated adaptation finance within a broader perspective of global political economy. In Table 1, the four research questions operationalize this framework by examining the visibility of vulnerability, the positioning of ODA within adaptation discourse, the governance and institutional conditions necessary for mainstreaming, and the degree to which SDGs are integrated into climate finance debates. These combined contributions address the fragmented representation of vulnerability in the literature and offer a systematic basis for assessing how adaptation-oriented development assistance has been mainstreamed in scholarly and policy domains.

Table 1. Mapping of research questions, core concepts, and analytical goals

Research Questions (RQs)	Key Concepts	Analytical Goals
RQ1: To what extent is the theme of adaptation finance addressed in the climate finance literature alongside the concepts of adaptive capacity and vulnerability?	Adaptation Finance, Adaptive Capacity, Vulnerability	To examine the representation of vulnerability-based needs in the literature and the discursive reflection of the mainstreaming principle.
RQ2: How are ODA and the broader development finance framework as represented in the literature in the context of climate adaptation?	ODA, Development Finance Framework	To investigate the sensitivity of ODA allocations to vulnerability and their alignment with the mainstreaming approach.
RQ3: Which structural themes are the concepts of governance, institutional gap, and policy coherence associated with in the literature on climate adaptation policies?	Governance, Institutional Gap, Policy Coherence	To analyze the role of governance structures and local capacity; to evaluate preconditions within the contexts of SDG 13/17/14.
RQ4: To what extent are SDG 13, SDG 17, and the broader concept of sustainable development addressed in an integrated manner within the climate finance literature?	SDG 13, SDG 17, Sustainable Development	To investigate the integration of adaptation aid into multi-level policy processes and its relationship with the coherence of SDGs 17 and 13.

These research questions aim to map the structural and conceptual contours of the literature while assessing how vulnerability-centered approaches in development finance at the intersection of adaptation, development assistance, and sustainability are reflected in academic discourse within the broader climate policy context. The proposed discussion simultaneously examined the vulnerability-related themes revealed through bibliometric analysis and the trends observed in the data from OECD Development Assistance Committee (DAC), so as to evaluate whether vulnerability-sensitive financing has been coherently mainstreamed.

3. Methodological Strategy

This section systematically presents the methodological approach, data collection and analysis procedures, and the resulting findings. The study aims to conceptually examine multiple interlinked policy domains including climate finance, adaptive capacity, development assistance, governance, and the SDGs, as an integrated whole. To this end, a theory-informed bibliometric review approach has been adopted, with the goal of analyzing thematic trends rooted in vulnerability and institutional capacity debates within the climate finance literature.

3.1 Theoretical-Driven Bibliometric Design and Research Method

This study was intentionally designed as a theory-driven conceptual mapping rather than a scope-maximizing literature search. Accordingly, the objective was not to retrieve all climate-finance publications but to map the academic visibility and interconnections of core concepts derived from Adger (2006)'s vulnerability–resilience framework. For this reason, a single integrated query linking five predefined conceptual clusters (i.e., adaptation finance, adaptive capacity, vulnerability, ODA, and governance/SDGs) provided stronger conceptual coherence than fragmented multiple searches. This was, therefore, not a scoping review but a theory-informed mapping exercise, aligned with the deductive logic of the conceptual framework.

The research design integrated bibliographic coupling and thematic clustering as the two core bibliometric techniques, aligned with the four theory-driven research questions guiding this study. Using VOSviewer and the Bibliometrix *R* package, the analysis examined keyword co-occurrence patterns, citation networks, and co-authorship structures. These tools, when adopted together, enable the identification of dominant thematic clusters and facilitate an assessment of how scholarly production aligns with broader sustainable development narratives (Donthu et al., 2021).

While bibliometric analysis provided quantitative structural mapping, the scoping review contributed qualitative analytical depth. Following the structured yet flexible framework developed by Arksey & O'Malley (2005), the scoping component captured thematic diversity across the literature and highlighted institutional orientations shaping academic debates. The integration of both methods strengthened the theory-informed interpretation in the study, rendering not only descriptive review but also conceptual interpretations to map how vulnerability, adaptive capacity, development finance, governance, and the SDGs intersect in the academic domain.

In this study, a single and conceptually integrated search query was employed. This strategy, increasingly common in structured scoping reviews and theory-informed literature analyses (Arksey & O'Malley, 2005; Boell & Cecez-Kecmanovic, 2015; Tricco et al., 2018), preserves conceptual coherence that often weakens when multiple fragmented search strings are applied. Although multi-query strategies may broaden coverage, they frequently dilute theoretical alignment and lead to thematic fragmentation. In contrast, the single-query approach maintains the conceptual integrity necessary for applying a vulnerability-based analytical framework (Gough et al., 2017). This approach enables the empirical operationalization of Adger (2006)'s vulnerability framework by systematically capturing the intersections among core concepts such as adaptation finance, adaptive capacity, vulnerability, ODA, governance, and policy coherence. Rather than treating these domains separately, the integrated query maps their overlapping thematic spaces within the scholarly literature (Grant & Booth, 2009).

Search string used in the analysis involved:

("climate finance" OR "green finance" OR "adaptation finance") AND
("adaptive capacity" OR "resilience" OR "vulnerability") AND
("official development assistance" OR "ODA" OR "development finance") AND
("governance" OR "institutional gap" OR "policy coherence") AND
("SDG 13" OR "SDG 17" OR "sustainable development")

This structure aligns with the principle of cross-sectoral conceptual integration (Boell & Cecez-Kecmanovic, 2015), gathering disciplines such as climate policy, development economics, environmental governance, and sustainability science under a unified conceptual framework.

To ensure transparency, the complete query string used in Scopus and WoS searches was reported directly in the main text, as recommended by the best practice in bibliometric guidelines. While Scopus returned 376 documents, WoS produced only a single relevant record. Rather than treating this discrepancy as a limitation, the study interpreted it as an empirical finding: adaptation-oriented and vulnerability-based research appeared structurally under-represented in certain indexing systems, particularly WoS, which had narrower coverage in development and governance-oriented journals (Mongeon & Paul-Hus, 2016). This reinforced the need for a theory-anchored search strategy, as conceptual clusters linked to vulnerability and ODA were concentrated disproportionately in Scopus.

The literature search was conducted across both the Scopus and Web of Science (WoS) databases using the unified conceptual query developed for this study. When applied with the "All Fields" option, the search yielded 376 records in Scopus but only a single record in WoS. Given the substantially broader coverage of Scopus in the social sciences, development studies, and governance-related fields, the bibliometric analysis was carried out using Scopus data.

Importantly, this discrepancy was not treated as a methodological limitation but an empirical finding in itself. Prior studies demonstrated that WoS provided stronger representation in natural and environmental sciences while offering comparatively narrow coverage in areas, such as development cooperation, institutional governance, and public policy (Mongeon & Paul-Hus, 2016). The near absence of vulnerability-oriented adaptation finance in WoS therefore indicated a structural marginalization of this research domain within major academic indexing systems.

Accordingly, the Scopus–WoS asymmetry was incorporated into the gap analysis of this study. It highlighted that even at the level of academic indexing, adaptation finance framed through vulnerability, governance, and ODA

remained underrepresented, reinforcing the need for a theory-driven conceptual mapping rather than a purely exhaustive search strategy.

3.2 Data Collection and Process of Analysis

The literature search was conducted in Scopus and WoS using the conceptually integrated single-query structure outlined in Section 3.1. This unified query derived from the five theoretical clusters (i.e., adaptation finance, vulnerability-adaptive capacity, ODA, governance, and SDGs) was applied on May 16, 2025 with the "All Fields" option in both databases, to ensure the broadest coverage of titles, abstracts, keywords, institutional affiliations, and cited references. This approach ensured full alignment between the retrieval procedure and the theoretical orientation in this study.

The marked disparity between Scopus ($N = 376$) and WoS ($N = 1$) was not interpreted as a methodological flaw, but rather as an empirical finding. Prior research showed that WoS had narrower coverage in development studies, social sciences, governance, and non-English scholarship (Mongeon & Paul-Hus, 2016). These domains corresponded closely to the conceptual core of adaptation finance and ODA. Accordingly, the WoS gap reflected a structural indexing limitation, not an inadequacy in the search design.

To mitigate the reduced bibliometric visibility of policy-oriented knowledge, the study incorporated policy datasets from OECD DAC and institutional reports (see Section 3.3). This integrative strategy strengthened the empirical grounding of the analysis, enabling the study to capture real-world patterns of development finance and policy coherence, which were essential for interpreting adaptation finance despite database asymmetries.

Bibliometric analysis was conducted using VOSviewer (van Eck & Waltman, 2010) and Bibliometrix (Aria & Cuccurullo, 2017), the two complementary tools widely used in structured literature reviews. VOSviewer enabled advanced mapping of co-occurrence networks, thematic proximity, and conceptual clusters, while Bibliometrix provided statistical evaluation of publication trends, authorship structures, and knowledge domains (Donthu et al., 2021). Data extraction, cleaning, and inclusion procedures followed the PRISMA extension for scoping reviews (PRISMA-ScR) guidelines (Tricco et al., 2018), to ensure methodological transparency and reproducibility.

Although bibliometric reviews might use additional techniques such as co-citation or historiographic mapping, the present study prioritized conceptual clustering and structural mapping, to be consistent with the deductive and theory-driven framework. This allowed us to examine how adaptation finance, vulnerability, adaptive capacity, ODA, and governance intersected in scholarly discourse, in line with the analytical logic of Adger (2006)'s vulnerability framework.

3.3 Findings

This section presents the empirical findings derived from the integration of theory-driven bibliometric analysis and OECD DAC data. The results are structured to first visualize the conceptual evolution of the literature and subsequently juxtapose these academic patterns with real-world trends of financial allocation.

3.3.1 Overview of integrated bibliometric and ODA findings

The bibliometric analyses conducted to address the research questions in this study drew on visualizations and statistical output generated using VOSviewer (version 1.6.20) and Bibliometrix. The analyses were evaluated in terms of conceptual clustering, thematic intensity, and temporal evolution within the climate finance literature. Figure 2 illustrates the annual distribution of academic publications between 2009 and 2025, based on 376 records retrieved from the Scopus database, encompassing abstracts, titles, keywords, authors, and institutional affiliations. The year 2009 was selected as the starting point of the analysis, as it marked the period when themes of academic visibility of climate finance, development assistance, governance, and sustainability began to consolidate in the literature.

The publication trends depicted in Figure 2 reveal a pronounced increase in scholarly output between 2009 and 2025, with a particularly sharp rise in the last five years. This growth reflects not only thematic intensification but also conceptual maturation within the field, closely aligned with major global policy milestones such as the adoption of the SDGs in 2015, the entry into force of the Paris Agreement in 2016, and the launch of the European Green Deal in 2019. Previous studies documented similar surges in climate-related research following these policy shifts; these were interpreted as evidence of expanding interdisciplinarity and conceptual diversification in climate finance, green finance, and sustainable development research (Bautista-Puig et al., 2020; Bhatnagar & Sharma, 2022; Donthu et al., 2021; Muchiri et al., 2022).

Beyond publication counts, Table 2 summarizes the general bibliometric indicators of the analyzed literature, to demonstrate the robustness and relevance of the dataset. An annual growth rate of 18.44% signalled a rapidly expanding research domain, while an average of 16.2 citations per document reflected strong scholarly visibility. The relatively young average document age (3.28 years) indicated the dynamism and policy relevance of the field. Moreover, an international co-authorship rate of 34.84% highlighted substantial global and interdisciplinary

collaboration. Taken together, these indicators confirmed that the dataset provided a conceptually rich and empirically sound foundation for examining the evolving structure of climate finance research.

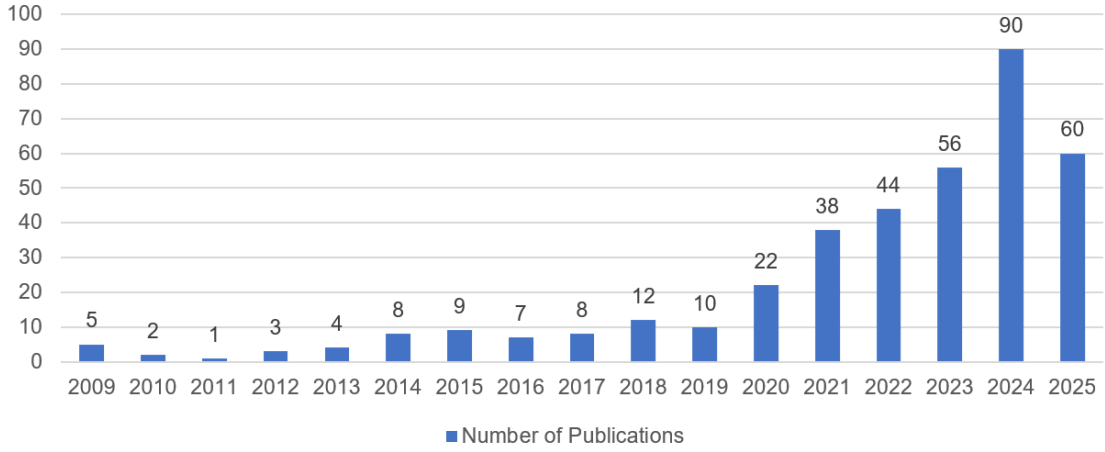

Figure 2. Annual distribution of publications from 2009 to 2025

Table 2. General bibliometric indicators of the analyzed literature from 2009 to 2025

Indicator	Value	Indicator	Value
Time span	2009–2025	Number of publisher keywords	511
Number of sources	263	Number of author keywords	1027
Number of documents	376	Number of authors	975
Annual growth rate (%)	18.44	Authors of single-authored documents	110
Average document age (years)	3.8	Average co-authors per document	2.9
Average citations per document	16.2	International co-authorship rate (%)	34.84

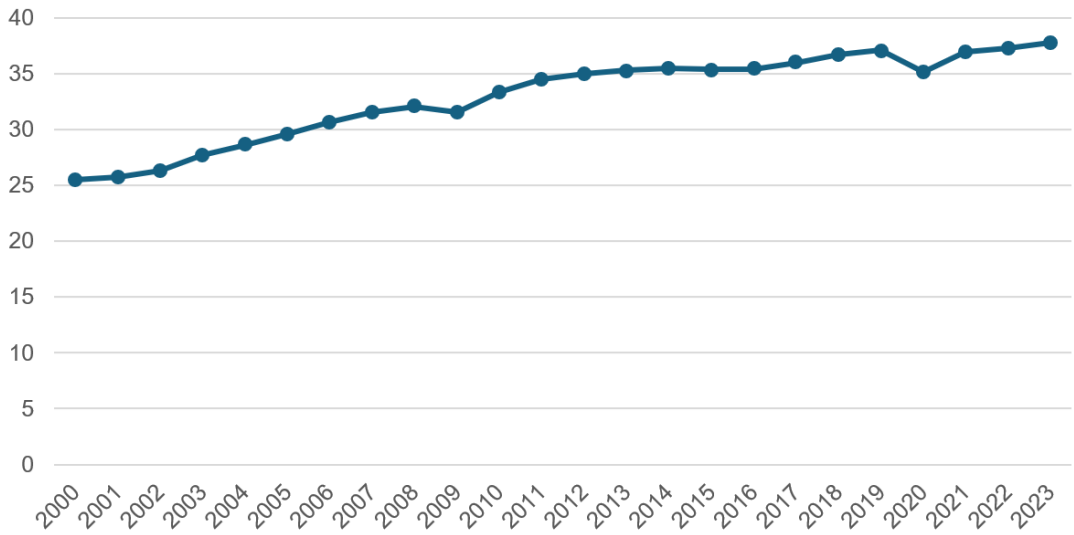

Figure 3. Global annual carbon dioxide emissions (Billion Tons)
Source: Hannah & Roser, 2020.

Building on these descriptive bibliometric characteristics, the analysis moved beyond publication trends and summary indicators to examine the conceptual structure of the literature. The evolving publication landscape illustrated a reciprocal relationship between the production of scientific knowledge and global climate policy, thus underscoring how academic research both responded to and shaped climate governance agendas over time. This transition enabled a systematic exploration of how key concepts such as adaptation finance, vulnerability, governance, and policy coherence interacted within the thematic architecture of the field, thereby directly supporting the conceptual and temporal coherence of the research questions in this study.

Building on the theoretical framework and bibliometric evidence presented in the previous sections, this part of the analysis explicitly examined whether the conceptual prominence of vulnerability-oriented adaptation finance

in the academic literature was reflected in development practice, with a specific focus on ODA-based policy patterns. The bibliometric analysis demonstrated that concepts such as adaptive capacity, vulnerability, governance, and policy coherence occupied a central position in the academic discourse on climate finance. This section therefore investigated the extent to which these conceptual orientations were mirrored in actual development assistance mechanisms, while also assessing the alignment between academic trends and global policy agendas. In this context, persistently rising carbon emissions and global warming trends, as illustrated in Figure 3, underscored the limitations of mitigation-centered strategies and highlighted the growing structural necessity of adaptation-oriented finance, particularly for vulnerable regions (Hannah & Roser, 2020). Accordingly, the findings suggested that adaptation finance, ODA, and governance were increasingly framed in the literature as interrelated components within climate finance debates, directly corresponding to the research questions on adaptation finance (RQ1), development assistance allocation (RQ2), institutional structures (RQ3), and sustainability-oriented policy integration (RQ4).

3.3.2 Mapping adaptation and vulnerability in climate finance: Bibliometric and ODA perspectives

Drawing from the integrated bibliometric and contextual evidence presented in Section 3.3.1, this subsection moves beyond descriptive publication trends to examine the relational structure of key concepts within the climate finance literature. Specifically, to assess how adaptation, adaptive capacity, and vulnerability were positioned relative to one another and to broader development finance themes, a co-occurrence network analysis was employed. This approach enabled the identification of conceptual proximity, thematic clustering, and structural coherence by mapping relationships among core keywords, thereby translating complex bibliometric data into interpretable network structures (van Eck & Waltman, 2010).

Crucially, this method directly responded to the patterns identified in the preceding analysis: while publication growth and thematic diversification signalled increasing academic attention to vulnerability-oriented adaptation finance, it remained unclear whether these concepts functioned as integrated pillars of the literature or persisted as fragmented and peripheral themes. By visualizing co-occurrence patterns, the network analysis allowed the evaluation of whether vulnerability-related concepts were embedded within the core adaptation finance discourse or remained marginal, despite their growing visibility. This analytical step was therefore particularly suitable to RQ1 as it provided a structural basis for assessing conceptual mainstreaming, prior to examining alignment or misalignment with ODA allocation patterns in subsequent sections.

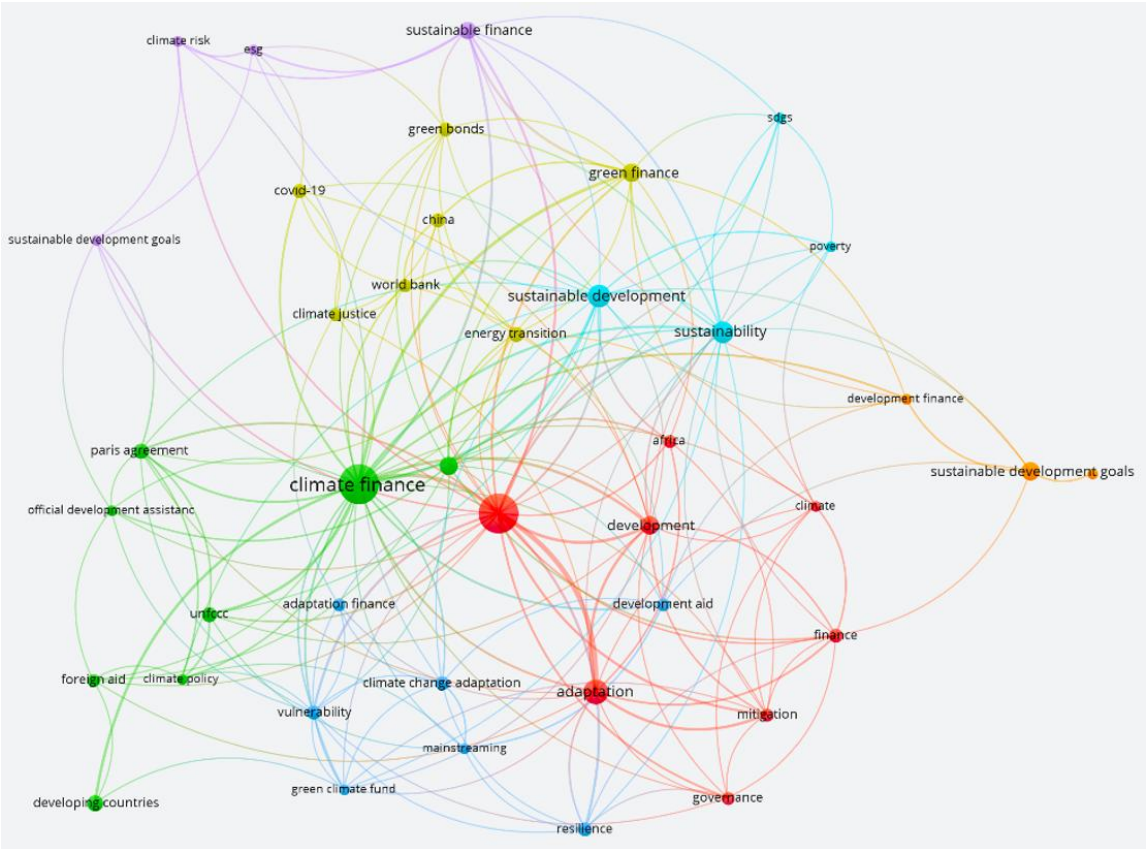

Figure 4. Conceptual co-occurrence network of author keywords in the literature

The analysis applied a minimum keyword frequency threshold, resulting in a filtered set of 41 keywords from an initial pool of 1,027 author keywords. The resulting network, illustrated in Figure 4, consisted of 7 clusters and 195 linkages. In this visualization, the node size reflected keyword frequency and relative importance, while link strength represented the intensity of co-occurrence. Clusters were generated based on content similarity and were displayed using distinct colors, allowing a structured interpretation of thematic groupings within the literature.

Figure 4 reveals a highly interconnected and multidimensional structure of the climate finance literature. The term climate finance occupies a central position in the network, exhibiting strong linkages with adaptation, vulnerability, and sustainable development. This configuration indicates that adaptation and vulnerability are not peripheral topics but constitute core conceptual anchors within the literature. The dense interconnections among these themes suggest that climate finance research has evolved into an interdisciplinary field encompassing environmental, governance-related, and socio-political dimensions, hence directly addressing the analytical scope of RQ1.

The red and green clusters in Figure 4 further highlight the structural integration between adaptation strategies and development-oriented financial mechanisms. Key concepts such as adaptation, finance, development, governance, and ODA exhibit both high frequency and strong interconnections, indicating that adaptation finance is closely associated with publicly supported funding structures and policy frameworks (Karg et al., 2025; Mantlana et al., 2025). This clustered structure reflects the conceptual foundation of adaptation finance as an instrument embedded within development finance and international aid architectures.

The blue cluster captures the social and structural dimensions of vulnerability through closely linked concepts such as vulnerability, adaptive capacity, and resilience. This cluster directly corresponds to the core focus of RQ1 by emphasizing adaptation capacity in the context of climate-related risks. Empirical studies underlined that vulnerability was strongly shaped by socio-economic constraints, including energy poverty and institutional fragility, which significantly limited adaptive capacity in developing countries (Nepal et al., 2025). At the same time, critical contributions questioned whether adaptation finance effectively reached the most vulnerable populations and highlighted persistent distributional concerns (Garschagen et al., 2021). These findings reinforced the central yet fragile positioning of vulnerability within the literature of adaptation finance.

While bibliometric evidence positioned adaptation and vulnerability at the conceptual core of climate finance research, actual financial allocations told a different story. Figure 5 illustrates bilateral ODA commitments by DAC members for climate change mitigation and adaptation between 2013 and 2022. Despite a gradual increase in adaptation-related funding, mitigation projects consistently receive a substantially larger share of total climate-related ODA. This persistent imbalance indicates that adaptation remains structurally underfunded relative to its prominence in the academic discourse.

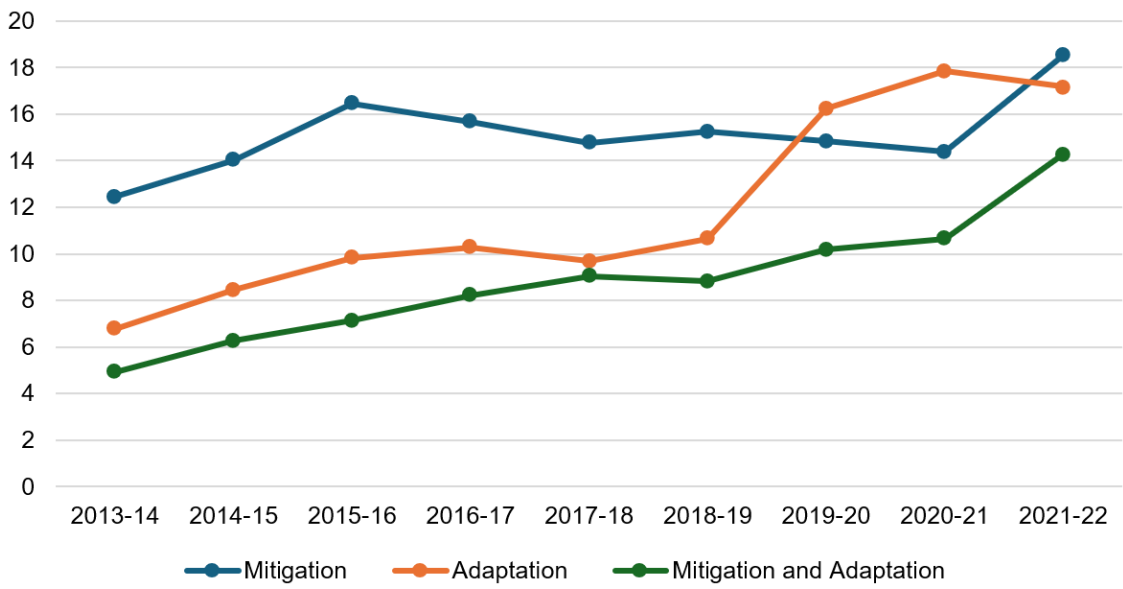

Figure 5. Bilateral Official Development Assistance commitments by DAC members for climate change adaptation and/or mitigation (Constant 2021 Prices, Billion USD)
Source: OECD, 2024.

The juxtaposition of bibliometric findings in Figure 4 and ODA allocation patterns in Figure 5 reveals a clear mismatch between conceptual emphasis and financial practice. Although adaptation, vulnerability, and adaptive capacity are positioned at the center of the academic literature, these themes remain underrepresented in donors' funding decisions. This disconnect is particularly problematic for developing countries, where adaptation needs

are most acute and closely tied to socio-economic vulnerability. Moreover, the observed imbalance is inconsistent with the objectives of SDG 13 and the principle of shared responsibility embedded in the Paris Agreement, which calls for a more equitable allocation of climate finance. In this sense, RQ1 highlights a structural disconnection between knowledge production and policy implementation, to form the central argument of this study.

3.3.3 Positioning core themes in the climate finance literature

Figure 6 presents the thematic map of the climate finance literature, positioning finance, development, and developing world as either motor or basic themes, depending on their centrality and density. Using the density–centrality framework proposed by Cobo et al. (2011), the map reveals that development-oriented finance concepts occupy a structurally central position within the literature, indicating their role as organizing pillars rather than peripheral topics. This configuration suggests that ODA-related discussions are conceptually embedded within broader debates on sustainable development and climate policy integration. Consistent with this structure, recent studies have emphasized the role of development finance as a coordinating mechanism linking climate objectives with institutional and policy frameworks (Karg et al., 2025; Sabbaghi, 2024). This conceptual configuration directly informs RQ2 by framing ODA not merely as a funding instrument, but as a core component of the development finance architecture.

Beyond core themes, Figure 6 also identifies a set of niche and basic themes that shape the operational boundaries of development finance. Concepts such as economic growth, empirical analysis, and carbon sequestration appear as niche themes, in order to reflect their concentration within specific subfields rather than the core literature. As noted by Favero & Hinkel (2024), such themes tended to emerge in sector-specific or technical discussions, particularly in relation to nature-based solutions and mitigation-oriented financing. In contrast, themes like environmental policy and adaptive management were positioned as basic themes, indicating their strong conceptual linkage to the core despite relatively lower thematic density. This pattern suggested that while ODA-related research recognized sectoral differentiation, integration across these domains remained partial (Mantlana et al., 2025).

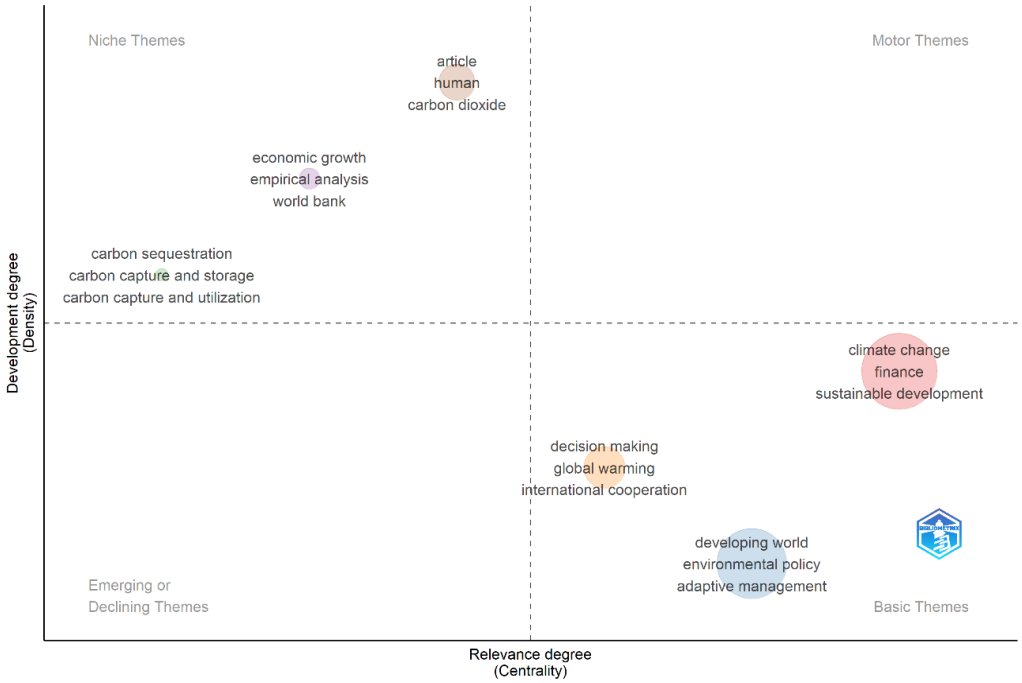

Figure 6. Thematic map

Figure 7 shifts the analysis from conceptual positioning to commitment patterns by illustrating ODA contributions as a share of gross national income (GNI) in the donor countries. The figure highlights substantial cross-country variation in meeting the SDG 17.2 target of allocating 0.7% of GNI to ODA. While a limited number of donors exceeded this benchmark, the majority remained below the target, thus underscoring the persistent commitment gaps in development finance (Our World in Data, 2025). When interpreted alongside the bibliometric emphasis on policy coherence and institutional responsibility, this distribution revealed a disconnect between normative commitments and actual resource allocation, particularly affecting vulnerable countries with high adaptation needs (Bergsvik et al., 2024). This commitment gap contrasted directly with the conceptual centrality of development finance, as identified in the thematic structure of the literature in Figure 6.

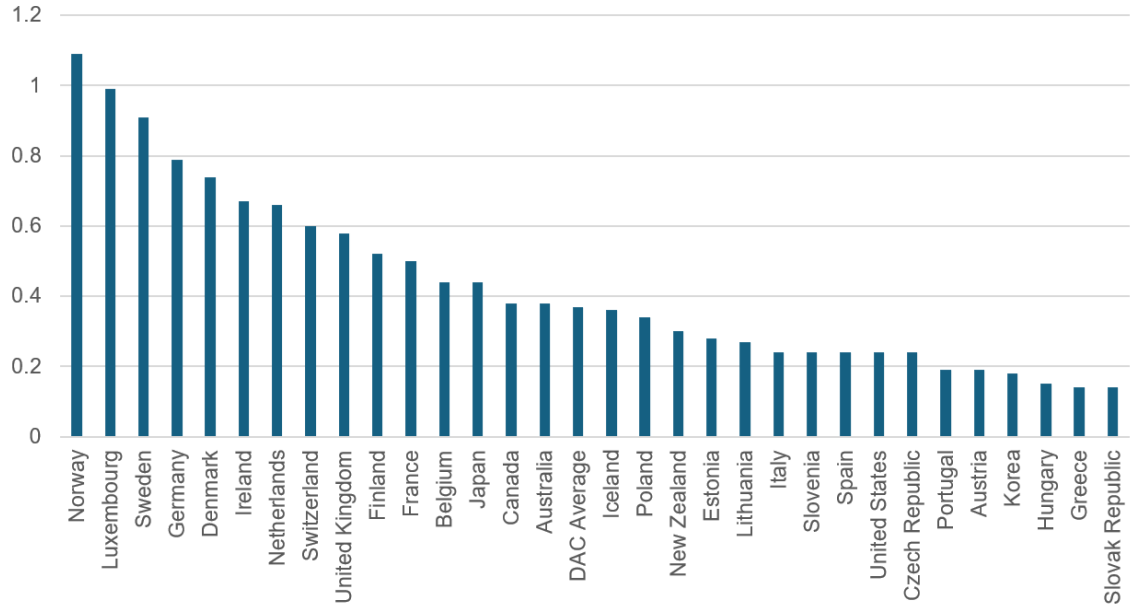

Figure 7. Official development assistance to developing countries (Percentage of the gross national ıncome of donor countries, 2023).
Source: Our World in Data, 2025.

Figure 8 further contextualizes these commitment gaps by illustrating the evolution of total ODA volumes and the composition of donors between 2000 and 2022. The data shows that while overall ODA volumes nearly tripled during this period, DAC countries continue to dominate contributions, with multilateral organizations gaining an increasing share (OECD, 2025). This trend aligns with the bibliometric prominence of concepts such as international cooperation and governance, yet it also raises questions regarding the effectiveness of volume growth in addressing vulnerability-based adaptation needs. As emphasized in the adaptation finance literature, increases in aggregate funding do not necessarily translate into improved alignment with development priorities unless accompanied by institutional and policy coherence (Mantlana et al., 2025; Nepal et al., 2025).

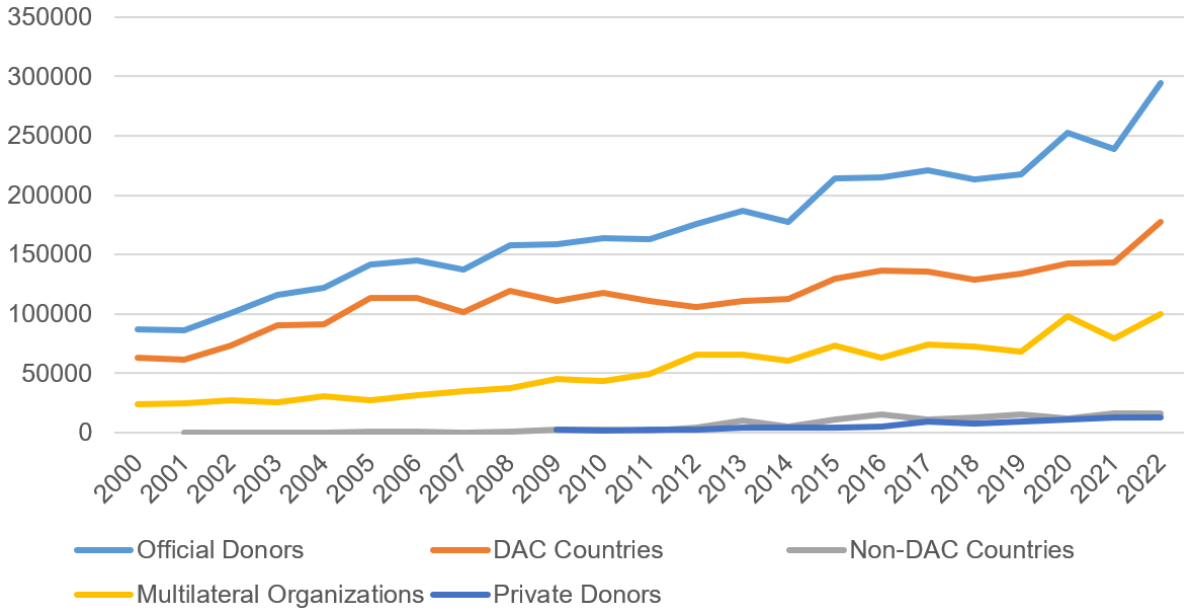

Figure 8. Official development assistance by donors (Million USD)
Note: Created by the authors based on data from the "Aid (ODA) by sector and donor [DAC5]" dataset available in the OECD Data Explorer (OECD, 2025).

Taken together, Figures 6–8 indicate that while ODA occupies a central position in the conceptual structure of the climate finance literature, its real-world allocation patterns remain only partially aligned with priorities of

vulnerability-driven adaptation. This divergence underscores that increases in aggregate ODA volumes, in the absence of stronger institutional coordination and policy coherence, are insufficient to ensure effective mainstreaming of adaptation finance within the broader development finance framework.

3.3.4 Conceptual shifts, governance priorities, and emerging topics

Figure 9 presents the temporal evolution of key concepts in the climate finance literature, revealing a pronounced post-2020 surge in terms of governance approach, decision making, and international cooperation. This trend indicates that climate finance research has increasingly shifted toward institutional and coordination-oriented perspectives, thus emphasizing the role of governance structures in shaping effective policy outcomes. The rising visibility of these concepts suggests a growing recognition that financial flows alone are insufficient without robust institutional frameworks that enable policy coherence and multi-actor coordination (Cobo et al., 2011; Izzuddin & Nainggolan, 2025). In this respect, governance emerges not as a peripheral theme but as a central analytical lens for understanding the capacity of implementing mechanisms of climate finance.

The bibliometric prominence of governance-related concepts aligns closely with broader debates on institutional gaps and policy coherence in climate governance. As highlighted in the literature, multi-level decision-making processes and mechanisms of international cooperation have become increasingly central, particularly in the context of developing countries where constraints of institutional capacity remain significant (Cho & Ackom, 2025; Muslim, 2024). These findings reinforce the relevance of RQ3 by demonstrating that governance is not merely an abstract concern but a structural determinant influencing how priorities of climate finance are translated into actionable policies. The literature thus frames governance as a critical intermediary between financial commitments and development outcomes. This institutional gap echoes recent findings which have emphasized that governance failures rather than financing shortages alone often limit the effectiveness of adaptation-oriented climate finance, particularly in vulnerable regions (Favero & Hinkel, 2024; Mantlana et al., 2025).

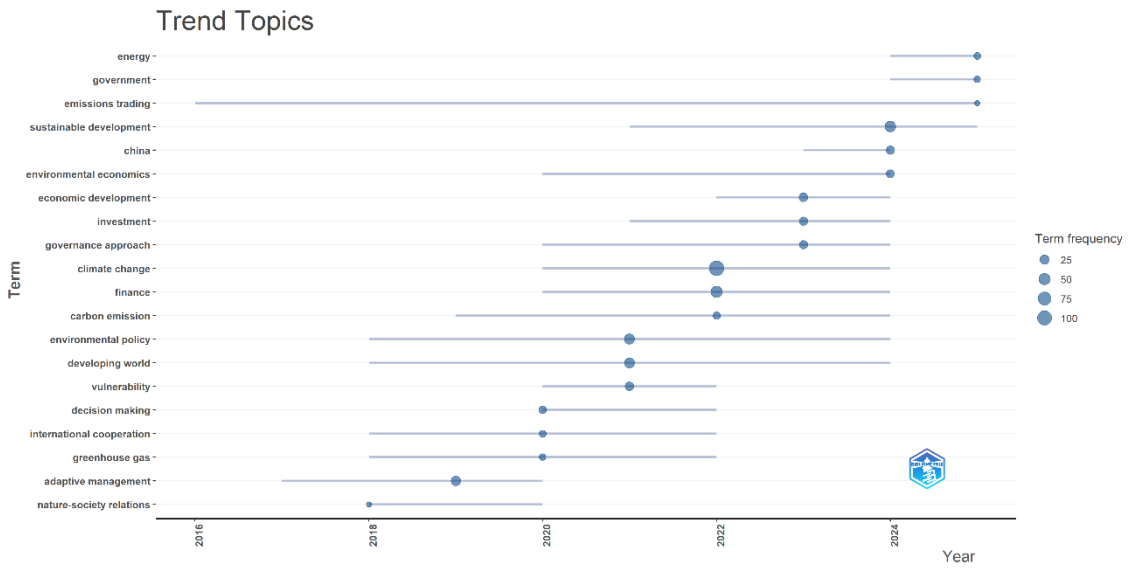

Figure 9. Trends of topics

Figure 10 shifts the analysis from conceptual trends to implementation patterns by illustrating the sectoral allocation of ODA between 2002 and 2023. Despite the bibliometric rise of governance and institutional coherence, ODA flows remain heavily concentrated in infrastructure- and mitigation-oriented sectors, such as energy and transport, while sectors more directly associated with governance strengthening and adaptation capacity receive comparatively limited support (OECD, 2025). This sectoral imbalance suggests a persistent disconnect between the governance-oriented priorities as emphasized in the literature and actual decisions of resource allocation, hence pointing to enduring institutional gaps in the implementation of policy.

When considered jointly, evidence from the combined bibliometric and ODA confirms that governance has become conceptually central yet remains institutionally underrepresented in the implementation of climate finance. While governance has gained conceptual prominence as a prerequisite for effective climate finance as shown in Figure 9, ODA allocation patterns in Figure 10 continue to underprioritize governance-strengthening and adaptation-related sectors. This structural gap substantiates RQ3 by demonstrating that policy coherence remains constrained not by a lack of conceptual clarity, but by institutional and implementation barriers. Addressing these gaps requires reframing climate finance not only as a volume-based commitment but as an institutionally embedded process aligned with governance capacities and development needs.

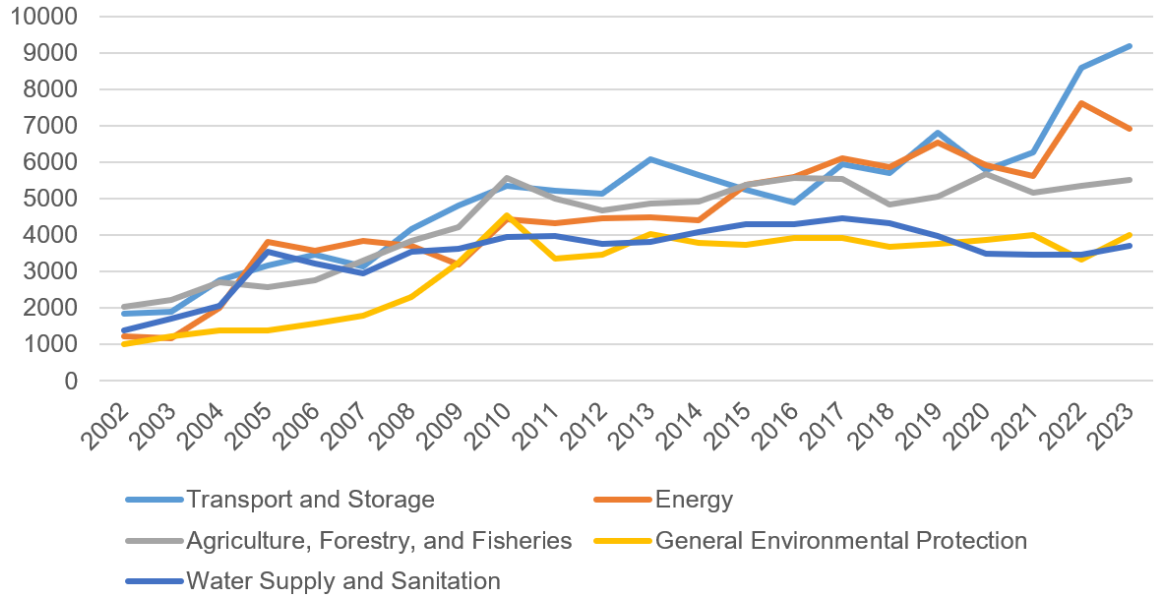

Figure 10. Official Development Assistance allocated to five sectors from 2002 till 2023 (Constant Prices, Million USD)

Note: Created by the authors based on the data from the "Aid (ODA) by sector and donor [DAC5]" dataset available in the OECD Data Explorer (OECD, 2025).

3.3.5 Tracing concepts: Thematic evolution analysis

Figure 11 illustrates the thematic evolution of concepts related to climate finance, revealing a marked post-2024 increase in the prominence of SDG-linked terms such as sustainable development, adaptive management, developing world, and decision making. This shift indicates that sustainability is no longer treated as a peripheral or aspirational concept but is increasingly integrated into the core analytical structure of the literature. The thematic continuity observed for climate change across periods, combined with the growing association of SDG-oriented concepts, suggests an evolving understanding of climate finance as a governance-embedded development process rather than a narrowly defined environmental or financial intervention (Cobo et al., 2011; Mantlana et al., 2025).

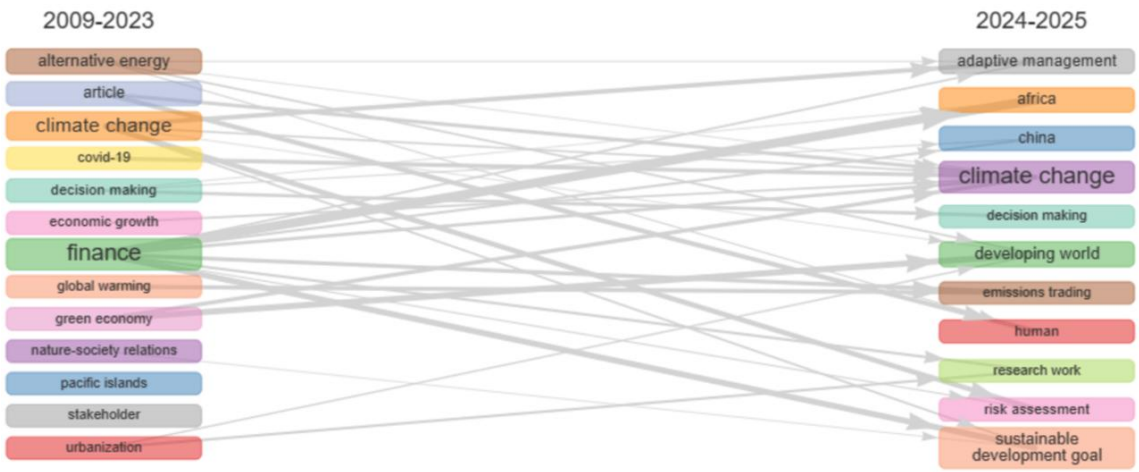

Figure 11. Thematic evolution analysis

Thematic evolution patterns further demonstrate a qualitative transformation in how sustainability is conceptualized. While earlier literature primarily associated finance with economic growth and green investment, recent contributions have increasingly positioned climate finance in a policy-steering and institutional coordination framework aligned with the SDGs. This transition reflects a broader recognition that sustainable development outcomes depend not only on financial volume but also on governance capacity, institutional coherence, and strategic alignment with development priorities (Galindo-Gutiérrez, 2024; Roy, 2024). In this light, SDGs emerge as structuring devices that connect climate objectives with development planning and institutional accountability, thus directly reinforcing the analytical focus of RQ4.

Despite the increasing bibliometric visibility of SDG-linked sustainability concepts, ODA allocation patterns reveal a partial and uneven operationalization of these priorities. As shown in Figure 12, energy-related ODA in 2023 remains heavily concentrated in large-scale production, transmission, and mitigation-oriented infrastructure, while adaptation-focused and vulnerability-reducing investments receive comparatively limited support (OECD, 2025). This distribution contrasts with the growing emphasis on resilience, adaptive capacity, and inclusive development in the literature, thus highlighting a persistent gap between sustainability discourse and financing practice. Such misalignment suggests that sustainability integration under SDG 13 and SDG 17 is constrained by dependencies on the sectoral and institutional path.

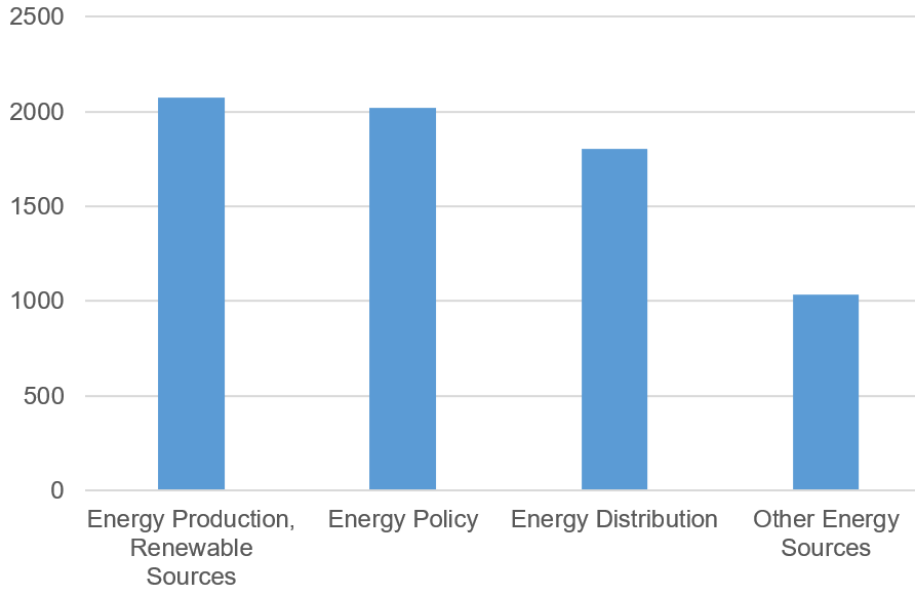

Figure 12. Sub-sectoral distribution of Official Development Assistance in the energy sector in 2023 (Constant Prices, Million USD)

Note: Created by the authors based on data from the "Rio Markers: Aid (ODA) activities targeting global environmental objectives" dataset available in the OECD Data Explorer (OECD, 2025).

Evidence from combining the bibliometric and ODA confirms that while sustainability has gained conceptual centrality in the research on climate finance, its translation into practices of development finance remains incomplete. The uneven incorporation of SDG-linked priorities into ODA allocation reflects structural and institutional barriers rather than a lack of conceptual clarity. In this context, RQ4 highlights that effective sustainability integration requires not only alignment with SDG narratives but also the institutionalization of vulnerability-based criteria, governance reforms, and mechanisms of cross-sectoral coordination. Without such embedding, sustainability risks remaining selectively implemented and unevenly distributed across sectors and regions (Halbherr et al., 2021; Mantlana et al., 2025).

4. Policy Alignment and Gaps

The findings presented in the previous section revealed that ODA-based climate finance exhibited certain structural patterns in terms of donor composition, burden sharing, thematic focus, and sectoral distribution. The contribution levels of donor countries fall significantly short of the SDG 17.2 targets. Within the sectoral allocation of ODA, large-scale infrastructure and mitigation-oriented investments are prioritized over adaptation-based needs. The distribution observed in the energy sector, in particular, indicates that mitigation-focused technical solutions receive greater financial priority. All these trends suggest that the conceptual clusters identified through bibliometric analysis such as adaptive capacity, vulnerability, ODA, resilience, and governance have limited reflection at the implementation level. However, this study argued that this misalignment was not merely an implementation failure but stemmed from deeper structural causes, including disciplinary silos and conflicting incentive structures between academia and policy.

This study employed core bibliometric analysis within an integrated framework, demonstrating that concepts such as vulnerability and governance are central in academic discourse. Studies by Garschagen & Doshi (2022) and Nepal et al. (2025) proved that these linkages were theoretically integrated through social vulnerability. However, a critical translation gap exists. While academic literature adopts complex and context-specific definitions of vulnerability (grounded in Adger's framework), institutions of development finance rely on easily

measurable and technical metrics (e.g., tons of CO_2 reduced). This creates a data-driven inertia: a lack of standardized and quantifiable indicators for adaptive capacity renders it difficult for donors to justify adaptation funding compared to mitigation projects, leading to the marginalization of vulnerability-based approaches in practice.

Findings from the thematic map analysis indicated that while finance and sustainable development were motor themes in literature, their operationalization was constrained. The literature typically framed ODA as a normative obligation for global equity (Malherbe & Oladejo, 2025). In contrast, the ODA data revealed a political economy constraint. Actual donor allocations are often driven by geopolitical interests and trade partnerships rather than pure vulnerability reduction. This explains the gap identified in RQ2: the academic ideal of needs-based allocation clashes with the interest-based allocation of realpolitik, hence preventing the seamless integration of development finance frameworks into policy.

The thematic evolution analysis revealed a shift towards decision making and SDGs from 2024–2025 (Mantlana et al., 2025; To, 2022); this suggested that finance was viewed as an integral component of comprehensive strategies (Favero & Hinkel, 2024; Galindo-Gutiérrez, 2024). However, this holistic academic view faced disciplinary silos at the implementation level. While sustainability science integrates ecological and social goals (SDGs 13 and 17), the operational guidelines of institutions related to development finance are often compartmentalized into specific sectors (e.g., energy and transport) without cross-cutting mandates. Consequently, the SDG-oriented policies identified in the trend analysis (Cho & Ackom, 2025; Muslim, 2024) remain rhetorically strong in academia but structurally weak in sectoral allocation.

In conclusion, the visibility of the integrated vulnerability-ODA approach remains highly limited in the academic literature indexed by WoS. The bibliometric analysis revealed that only one study addressed this tripartite structure in a comprehensive manner. Therefore, the study identified a conceptual misalignment (literature–policy mismatch). This mismatch persists because academic incentives favor novel theoretical contributions, while policy mechanisms prioritize risk-averse and quantifiable project output.

5. Structural Limits to Vulnerability-Based Adaptation Finance

The bibliometric analyses conducted in this study revealed that vulnerability-centered adaptation finance was a prominent thematic focus in academia yet remained limited in practice. This section critically examined the boundaries of the current policy architecture and proposed specific mechanisms to bridge these gaps. It demonstrated that adaptation finance was not merely a resource allocation issue but a matter of institutional design.

The increasing volume of adaptation finance enhances potential, but Jakob et al. (2014) argued that large-scale climate finance did not always produce positive outcomes; on the contrary, high-volume funding might trigger rent-seeking behavior and financial volatility (climate finance curse), particularly under weak governance. Similarly, Kalinowski (2024) illustrated that the dominant role of the private sector led to a drift away from adaptation goals. To bridge this gap, adaptation finance should move beyond volatile project-based funding. A counter-cyclical financing mechanism should be institutionalized within ODA frameworks, to ensure steady flow of funds to vulnerable countries even during global economic downturns, thereby mitigating the macroeconomic risks identified in the literature.

Allocation is often shaped by priorities of foreign policy rather than need. In their study focusing on Southeast Asia, Gupta & Saradhi (2025) demonstrated that major donors prioritized trade and security, thus turning adaptation finance into a matter of political economy based on the argument by Betzold & Weiler (2017). To address this distortion, the OECD DAC should co-develop new vulnerability-weighted allocation criteria with academic networks. This mechanism would require donors to report not just the volume of finance but also its vulnerability alignment, in order to create accountability for interest-based deviations.

Recent findings have indicated structural inequality in which the nations most at risk are excluded. According to Garschagen & Doshi (2022), only 53% of resources from Green Climate Fund (GCF) were directed to countries with the highest vulnerability. Similarly, Islam (2022) highlighted a Low Funding Trap in which the most vulnerable received disproportionately low support. Supporting this view, Rauniyar et al. (2025) revealed that adaptation-focused ODA did not linearly align with risk levels. Furthermore, adaptation readiness often excluded those with limited capacity (Jain & Bardhan, 2022), hence producing a double burden exacerbated by rising debts (Klusak et al., 2023) and weakening the very capacity required to access funds out of climate change (Callahan, 2025). This systemic barrier requires a shift from ex-ante conditionality to ex-post accountability. Instead of demanding institutional readiness before funding, international funds should deploy readiness-building grants that specifically target the administrative gaps which could prevent access.

Effectiveness depends on multi-level governance (Sherman et al., 2016), yet globalization often disconnects local knowledge from the implementation of policy (Berkes et al., 2008). Furthermore, recent insights from the International Monetary Fund (IMF) (Mitra et al., 2025) have highlighted that the absence of vulnerability indicators in macroeconomic statistics creates a problem of visibility. This points to a critical area for bridging the policy-academia gap and data integration. Academic researchers and international financial institutions should

collaborate to integrate vulnerability indices directly into national accounts and credit rating methodologies. Making vulnerability macro-economically visible is the only way to move adaptation from a peripheral environmental concern to a central economic mandate.

These findings underscore the need for a new governance architecture. The mismatch between academic theory and policy practice is not unsolvable but requires deliberate institutional reforms, involving moving from project-based to programmatic funding, integrating vulnerability data into economic planning, and aligning donor incentives with long-term resilience goals.

6. Conclusions

This study conceptually and empirically examined the structural disconnect between vulnerability-oriented adaptation discourse and actual ODA allocation patterns. The bibliometric analysis revealed that while concepts such as adaptive capacity, vulnerability, and governance had achieved conceptual centrality in academic literature, ODA data indicated that these priorities remained marginalized in practice. The core finding of this study was not merely the existence of a gap, but the identification of a structural misalignment where the normative expansion of vulnerability in theory failed to penetrate the operational logic of development finance.

This misalignment highlights a critical normative-analytical divide, while global frameworks (SDGs and Paris Agreement) have normatively mainstreamed vulnerability. Academic research has often failed to translate these soft concepts into hard and quantifiable metrics required by institutions of development finance. This divide empirically demonstrates that global debates about macroeconomic policy, often conducted in abstract terms regarding equity and resilience, face their most critical test of implementation within the architecture of adaptation finance. Consequently, this study identified adaptation finance not merely as a sub-sector of aid, but as the precise locus where these global debates were either operationalized or structurally blocked, thus leaving the structural vulnerabilities of the Global South as unaddressed systemic risks within the world economy.

To bridge this literature–policy gap, this study proposed a set of specific pathways for institutional reform, moving beyond generic calls for increased funding:

- Aligning volume with vulnerability: The OECD DAC should move beyond reporting aggregate ODA volumes and co-develop vulnerability-weighted allocation criteria with academic networks. This mechanism would require donors to demonstrate how their allocation explicitly targets the specific vulnerability indices of recipient nations, in order to create accountability for geopolitical deviations.
- Institutionalizing counter-cyclical support: To address the climate finance curse and volatility identified in the literature, international funds such as Green Climate Fund (GCF) should institutionalize counter-cyclical financing mechanisms. These instruments would ensure guaranteed liquidity for vulnerable nations during global economic downturns, to decouple adaptation finance from donors' economic cycles.
- Overcoming the capacity trap: The current criteria of adaptation readiness act as exclusionary barriers. Financial institutions should shift from ex-ante conditionality to ex-post accountability by deploying readiness-building grants. These grants should specifically target the administrative and data gaps in low-income countries that prevent them from accessing larger adaptation funds.
- Integrating data as a policy lever: Adaptation finance will remain invisible in macroeconomic planning unless systematically measured. The IMF and National Statistical Offices should integrate vulnerability indices directly into national accounts and credit rating methodologies. Making vulnerability macro-economically visible is the prerequisite for shifting adaptation from a peripheral environmental concern to a central economic mandate, recognizing that climate vulnerability in developing nations constitutes a material risk to global financial stability.

While providing a novel theoretical integration, this study acknowledged two key limitations. First, the restriction to peer-reviewed literature might limit the visibility of practice-oriented knowledge found in grey literature; future studies are strongly encouraged to triangulate findings with policy reports from United Nations (UN) or non-governmental organizations repositories. Second, the analysis of DAC data relied on descriptive structural alignments rather than causal modeling; subsequent research is encouraged to employ comparative and empirical modeling approaches to rigorously test the determinants of vulnerability-based adaptation finance.

In conclusion, bridging the gap between adaptation theory and financial practice requires a fundamental shift in institutional design. Future research should move beyond descriptive gap analysis to empirically test the impact of these proposed governance mechanisms. By integrating vulnerability not just as a theoretical concept but as a governing metric of finance, the global community could align ODA realities with the "leave no one behind" principle. Ultimately, this transition is essential to reframe adaptation finance not merely as an assistance to development, but as a structural pillar of a resilient and inclusive world economy.

Author Contributions

Conceptualization, P.Y.; methodology, software, validation, formal analysis, investigation, resources, data

curation, visualization, C.O.; writing—original draft preparation, writing—review and editing, supervision, P.Y. and C.O. All authors have read and agreed to the published version of the manuscript.

Data Availability

The bibliometric data used in this study are available from the Scopus (https://www.scopus.com) and Web of Science (https://www.webofscience.com) databases, yet both of which require a subscription for access. The ODA datasets are publicly available from the OECD Development Assistance Committee (DAC) database at https://data-explorer.oecd.org/. No new data was created in this study.

Conflicts of Interest

The authors declare no conflicts of interest.

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